voltage_divider_routed

Design Report

kicad-tools 0.23.0

Rev 1 | 2026-10-18 | jlcpcb

Board Summary

Property	Value
Layers	2 copper (F.Cu, B.Cu)
Footprints	4 (2 SMD, 2 THT, 0 other)
Nets	3
Traces	14 segments
Vias	3
Board Size	30.0 x 25.0 mm

Design Overview

Theory of Operation

Voltage Divider Test

Simple 2-resistor voltage divider

5V -> 2.5V (10k/10k)

Power Architecture

Power Rails: GND, VIN

ERC Status

Metric	Count
Errors	0
Warnings	0

Status: PASS

Schematic Overview

Schematic: voltage_divider

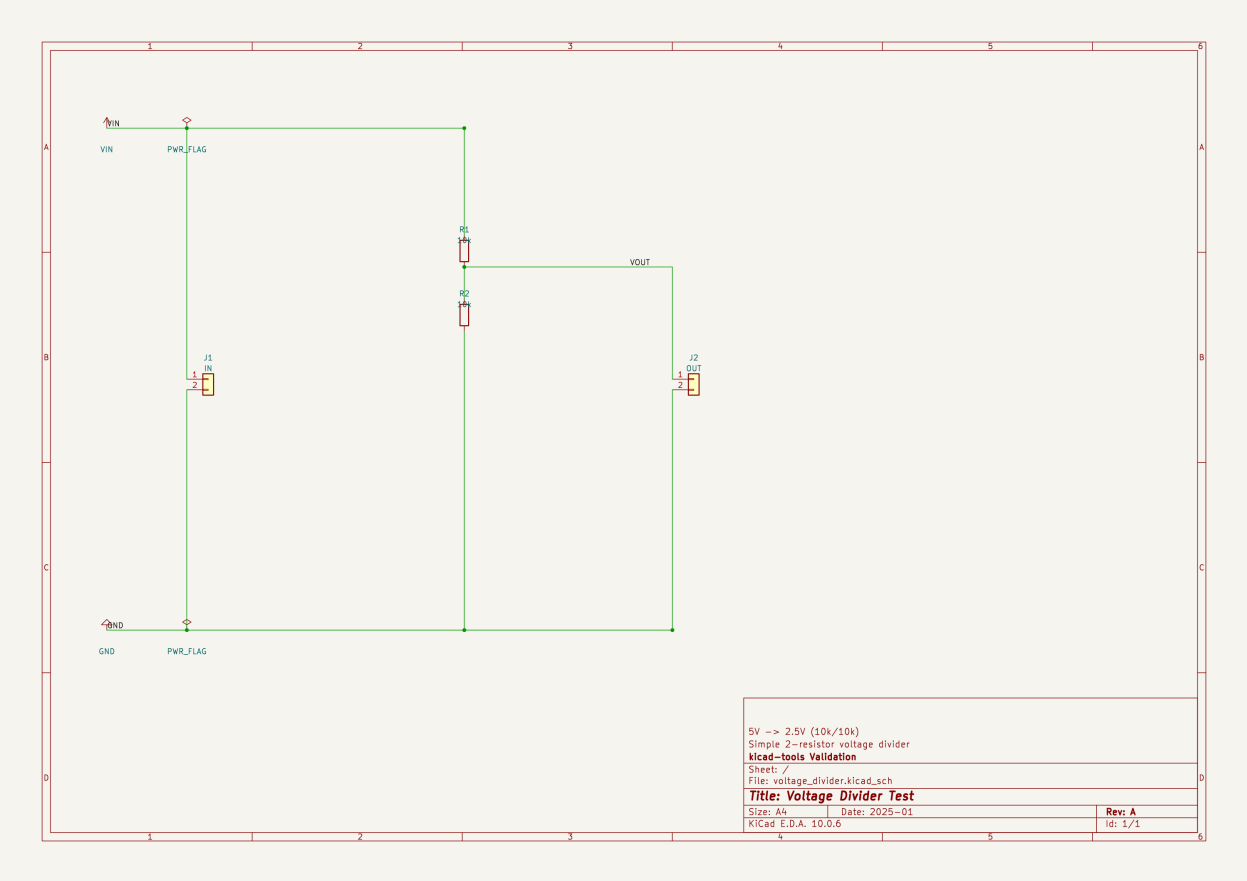


Figure 1: Schematic: voltage_divider

PCB Layout

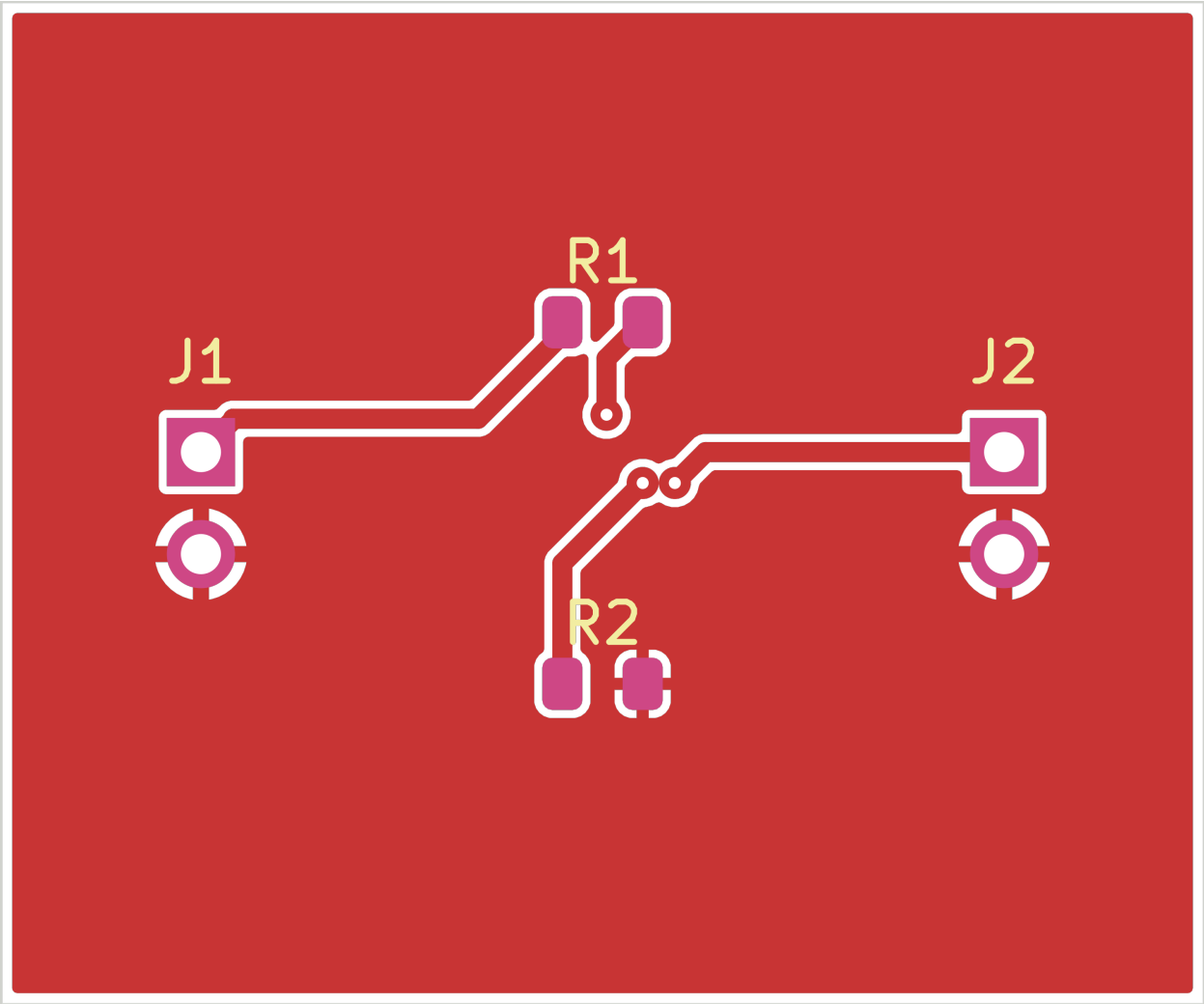


Figure 2: PCB Front

Copper

Assembly

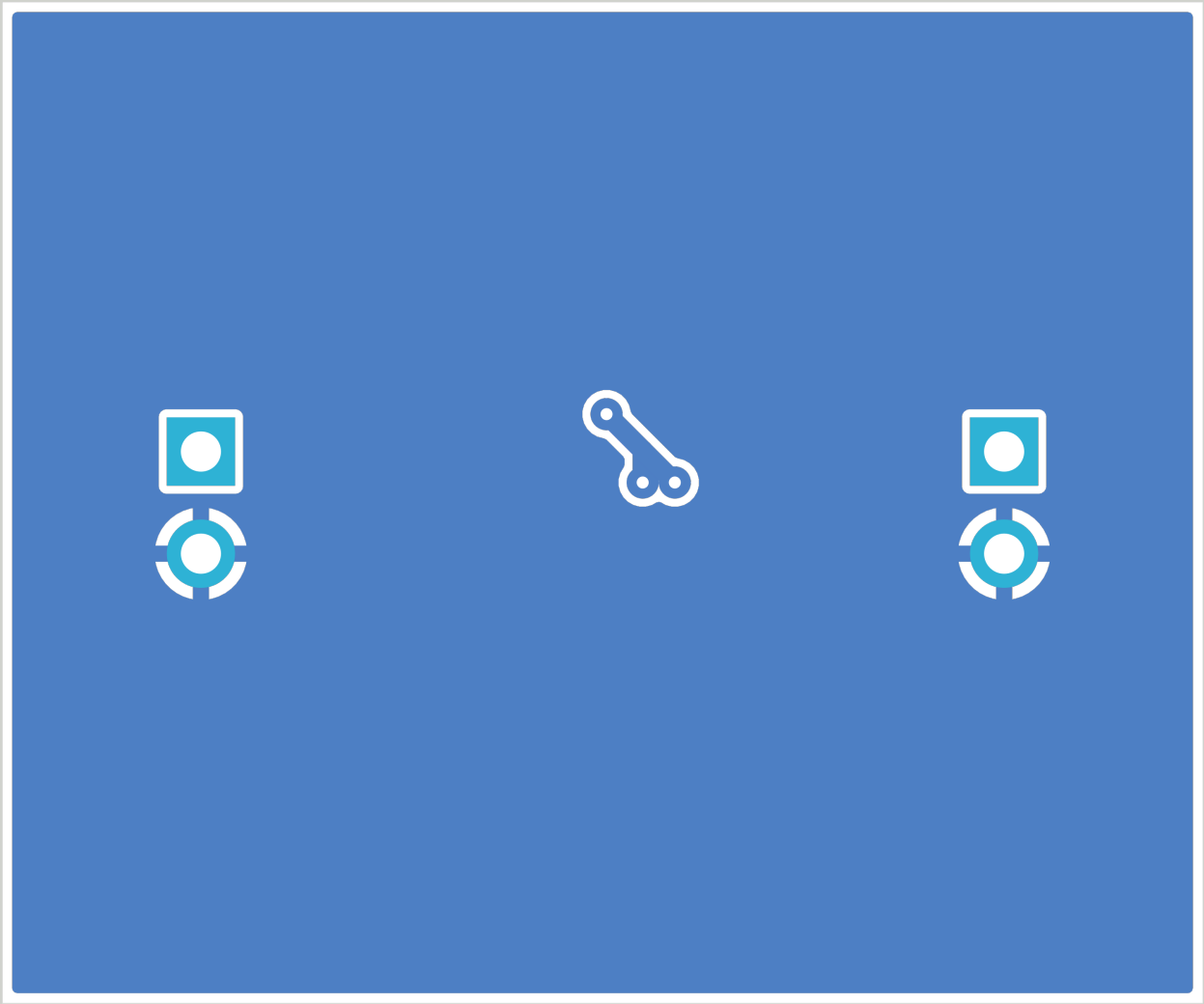


Figure 3: PCB Back

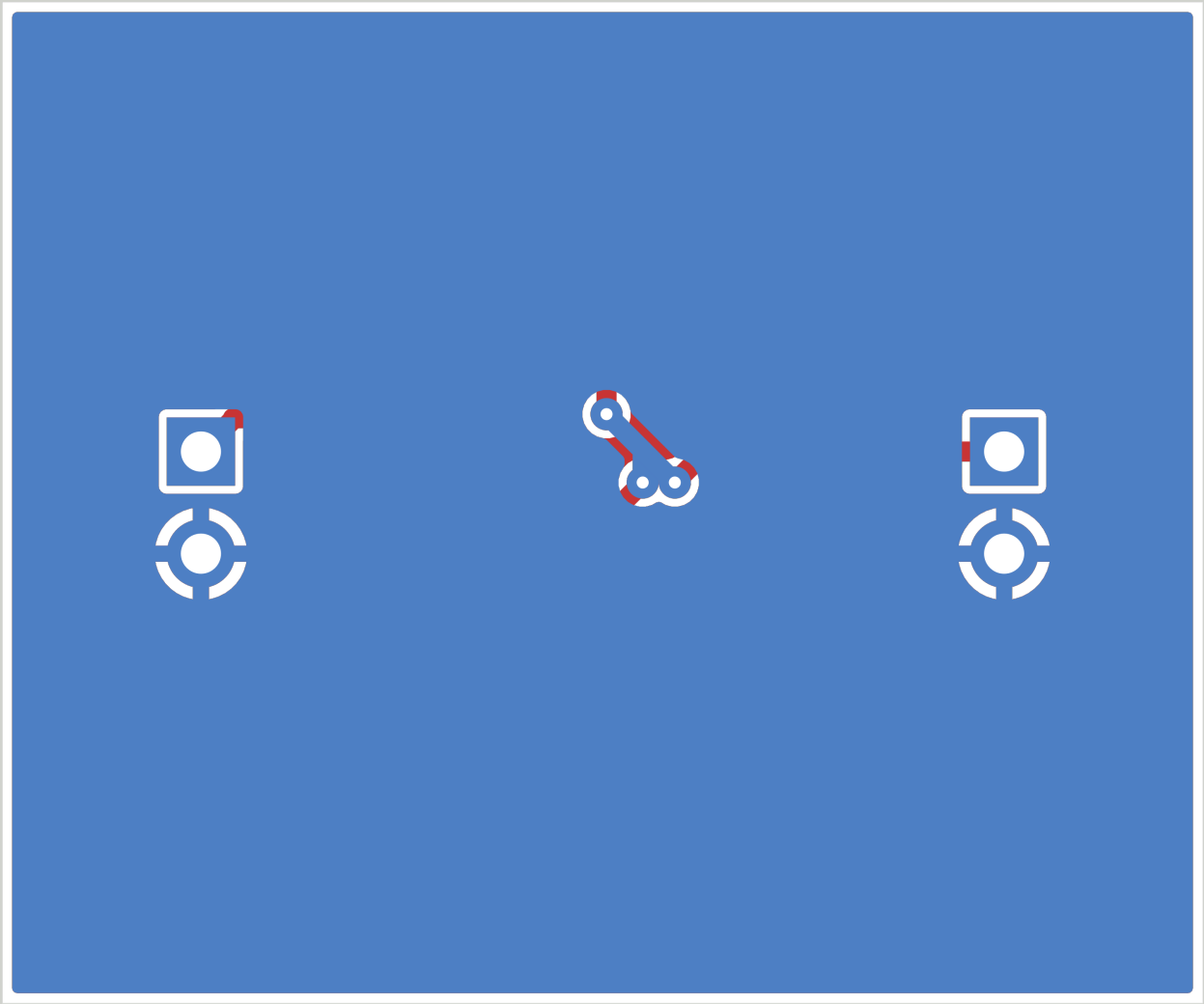


Figure 4: PCB Copper

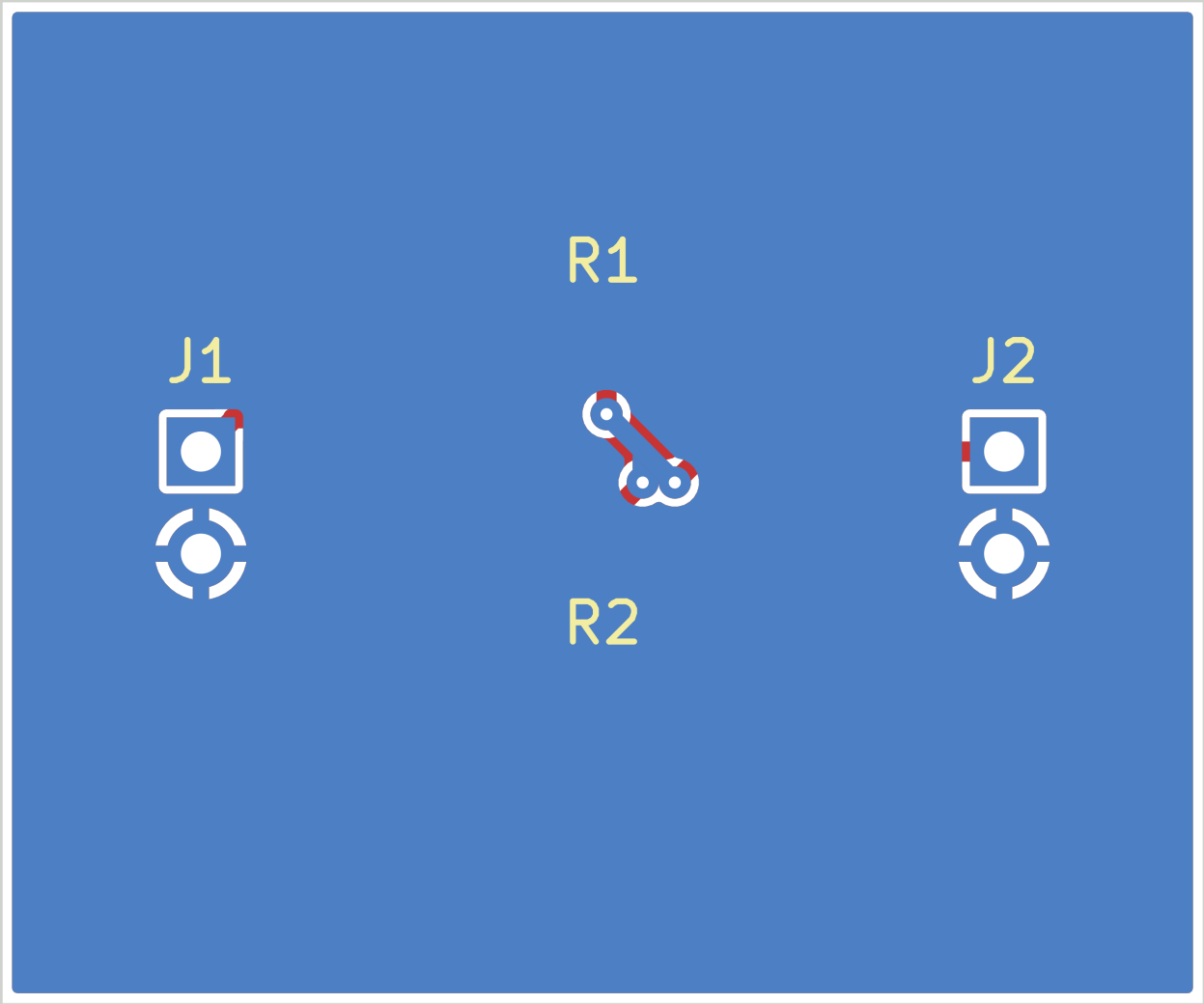


Figure 5: Assembly

Copper Layers

F.Cu

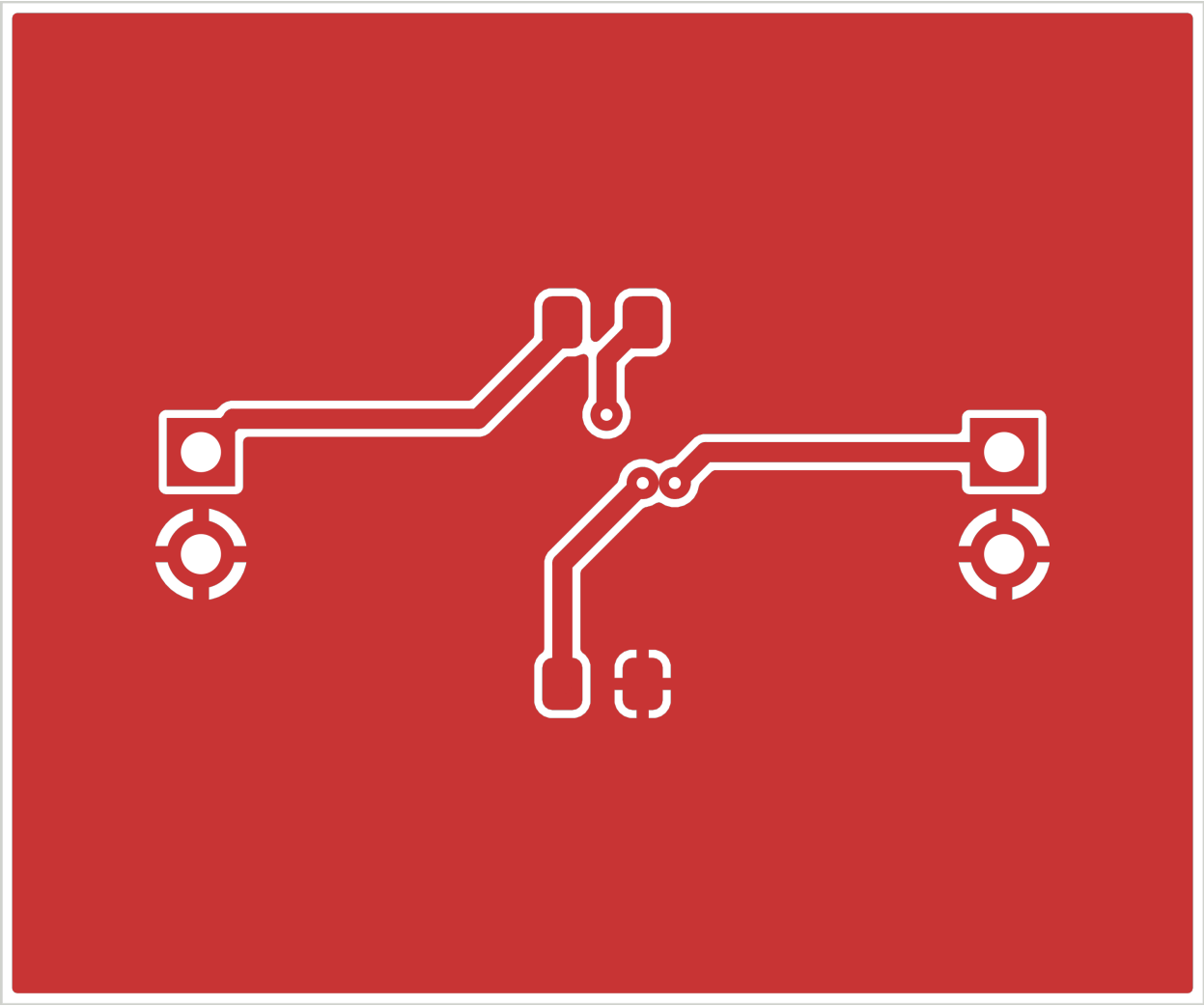


Figure 6: F.Cu

B.Cu

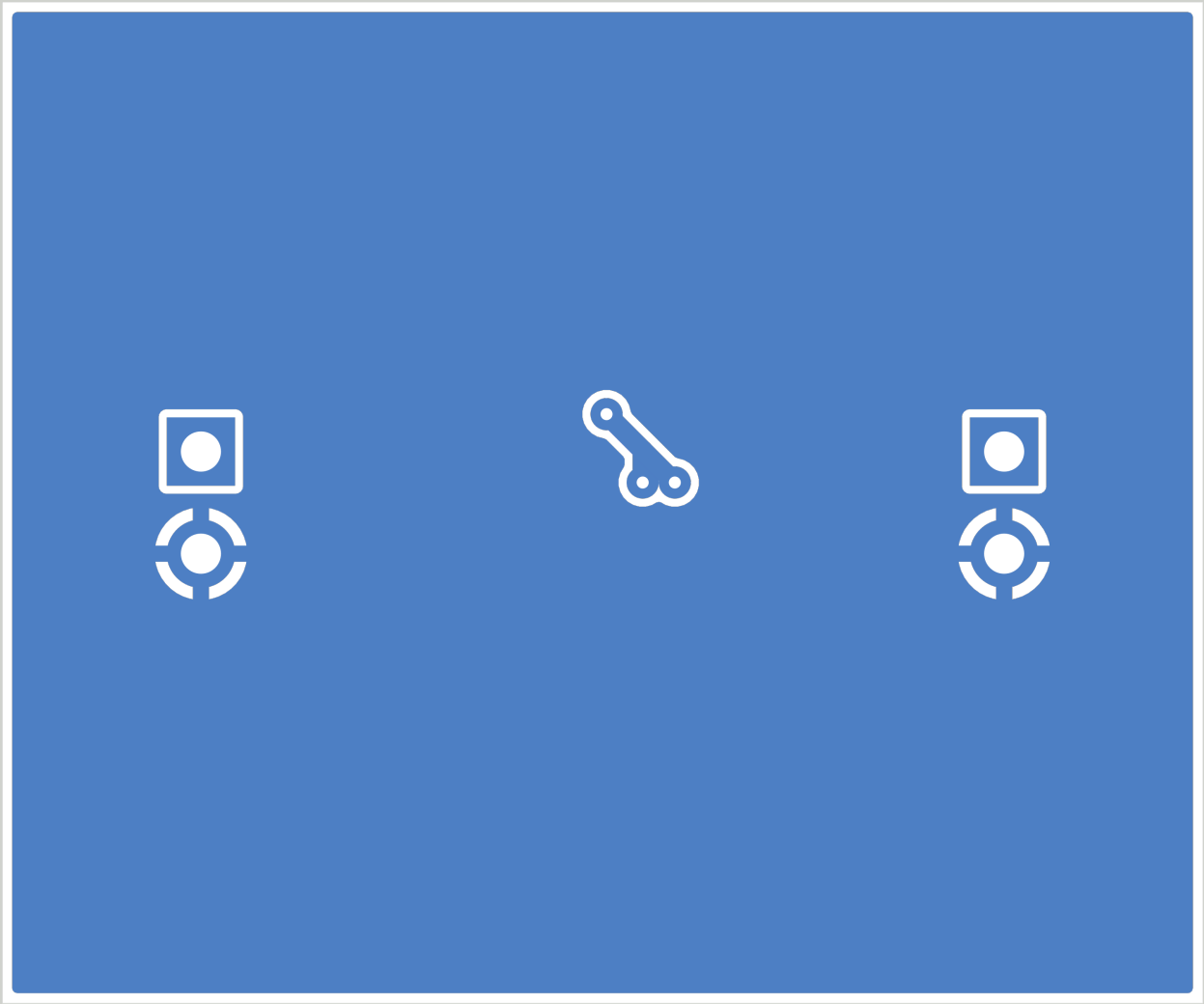


Figure 7: B.Cu

Bill of Materials

Value	Package	Qty	References
IN	PinHeader_1x02_P2.54mm_Vertical	1	J1
OUT	PinHeader_1x02_P2.54mm_Vertical	1	J2
10k	R_0805_2012Metric	2	R1, R2

DRC Status (`--mfr jlcpcb` rules)

Metric	Count
Errors	0
Errors: kct (`--mfr jlcpcb` rules)	0
Errors: kicad-cli (geometric)	0
Warnings	0
Blocking	0

Status: PASS

Manufacturing Readiness

Verdict: READY

Action Items

- [**OPTIONAL**] Refill zones to verify 1 zone-connected nets, then re-run kct audit

Routing Status

Metric	Value
Signal Net Completion	100.0% (2/2)
Overall Completion	100.0%
Complete Nets	3 / 3
Zone-Connected Nets	1
Incomplete Nets	0
Unconnected Pads	0

Zone-Connected Nets

- GND

Cost Estimate

Metric	Per Board (estimated)
PCB Fabrication	~0.55 USD
Components (estimated)	~0.21 USD
Assembly (estimated)	~1.93 USD
Total (estimated)	**~2.69 USD**
Batch Quantity	5
Batch Total (estimated)	~13.44 USD